

Business Outcomes and Psychological Difference: A Mathematical Analysis of Agreeableness and Income Disparity

1 Psychological Traits as Economic Variables

In labor economics and organizational psychology, personality traits are modeled as *latent individual differences* that systematically affect market outcomes. Let

$$A \in \mathbb{R}$$

denote an individual's standardized agreeableness score (mean 0, variance 1), as measured by Big Five psychometrics.

Empirically, agreeableness reflects tendencies toward cooperation, compliance, conflict avoidance, and prosocial behavior. These traits interact directly with negotiation behavior, promotion dynamics, and compensation structures in firms.

2 Income as a Function of Psychological Traits

Let annual income be modeled as

$$Y = f(A, X) + \varepsilon,$$

where:

- A = agreeableness
- X = vector of controls (education, experience, occupation, cognitive ability)
- ε = idiosyncratic noise

A firstorder approximation yields a linear earnings equation:

$$\log Y = \beta_0 + \beta_A A + \beta_X X + \varepsilon. \tag{1}$$

3 Magnitude of the Income Penalty

Empirical estimates place β_A in the range

$$\beta_A \approx -0.05 \text{ to } -0.10,$$

implying that a onestandarddeviation increase in agreeableness corresponds to a 5–10% *decrease in income*.

Thus, comparing two individuals identical in all respects except agreeableness:

$$\frac{Y_{\text{low } A}}{Y_{\text{high } A}} \approx e^{-\beta_A \Delta A}.$$

For $\Delta A = 2$ (one SD below vs one SD above the mean),

$$\frac{Y_{\text{low } A}}{Y_{\text{high } A}} \approx 1.10\text{--}1.22.$$

4 Mechanism: Business Interaction Model

This disparity arises from repeated strategic interactions. Let wage growth be modeled as:

$$\Delta Y_t = g(N_t, C_t, A),$$

where:

- N_t = negotiation frequency
- C_t = conflict tolerance
- A = agreeableness

Empirically one expects

$$\frac{\partial N_t}{\partial A} < 0, \quad \frac{\partial C_t}{\partial A} < 0,$$

which suppresses cumulative wage growth over time.

5 Income Distribution Conditional on Agreeableness

Let income conditional on agreeableness be lognormally distributed:

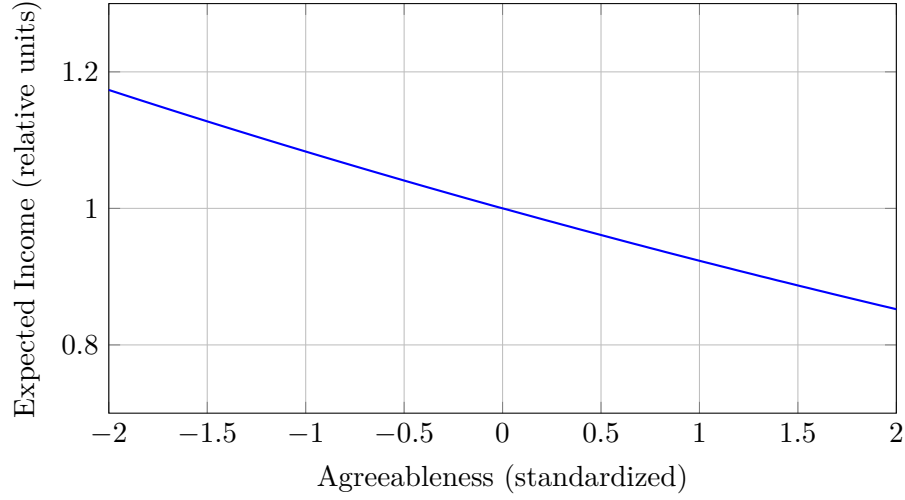
$$Y \mid A \sim \text{LogNormal}(\mu(A), \sigma^2), \quad \mu(A) = \mu_0 + \beta_A A.$$

Then expected income is:

$$\mathbb{E}[Y \mid A] = \exp\left(\mu_0 + \beta_A A + \frac{\sigma^2}{2}\right),$$

which is strictly decreasing in A when $\beta_A < 0$.

6 Graphical Representation



This curve represents:

$$\mathbb{E}[Y \mid A] \propto e^{\beta_A A}, \quad \beta_A = -0.08.$$

7 Interpretation for Business Systems

From a systems perspective:

- Firms reward *assertive surplus extraction*.
- Agreeableness increases cooperation but reduces bargaining leverage.
- Markets price outcomes, not intentions.

Hence, agreeableness acts as a *negative return personality factor* in competitive compensation environments.

8 Conclusion

Mathematically and empirically:

$$\boxed{\frac{\partial \mathbb{E}[Y]}{\partial A} < 0}$$

Agreeableness improves social harmony but is associated with a measurable income penalty in business environments governed by negotiation, competition, and asymmetric power.